

ECS Project Step 7 – Run ECS Task for Testing

To Begin with the Lab

Summary of the Lab

In this lab, the previously created **Amazon ECS Task Definition** was used to launch the first **Amazon ECS Fargate Task** inside an existing ECS cluster. During task creation, the task definition family, revision, desired task count, networking, and capacity provider strategy were configured. After the task entered the **Running** state, its public IP address was used to access the PHP web application. An image upload test confirmed that the application was working correctly by successfully storing the uploaded file in the configured Amazon S3 bucket. The lab also explained how capacity providers distribute tasks and how security group rules affect application accessibility.

Understanding the Lab

An **Amazon ECS Task** is a running instance of a Task Definition. The Task Definition acts as the blueprint, while the Task is the actual running container. Before a task can run, it must be launched inside an ECS Cluster.

When a task starts, ECS performs the following actions:

- Pulls the Docker image from Amazon ECR.
- Creates the required networking resources.
- Assigns the Task Role and Task Execution Role.
- Starts the container.
- Makes the application available through the assigned public IP address.

The lab also introduced **Capacity Provider Strategy**, which determines where ECS should run tasks. The **Base** value guarantees the minimum number of tasks on a specific capacity provider, while the **Weight** value distributes the remaining tasks proportionally among the available providers.

Example

Suppose you launch **10 tasks** with the following configuration:

- **FARGATE**: Base = 2, Weight = 1
- **FARGATE_SPOT**: Base = 0, Weight = 3

ECS first places **2 tasks** on FARGATE because of the Base value. The remaining **8 tasks** are divided according to the weight ratio **1:3**, resulting in **2 additional tasks on FARGATE** and **6 tasks on FARGATE_SPOT**. The final distribution becomes:

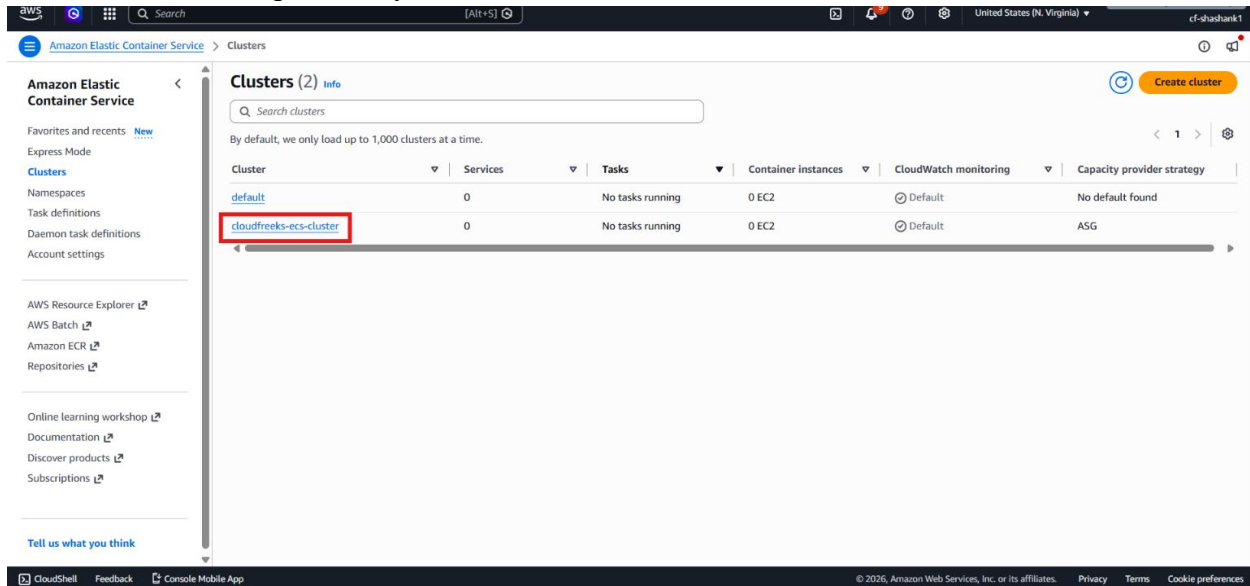
- FARGATE = 4 Tasks
- FARGATE_SPOT = 6 Tasks

For this lab, only **one Fargate task** was launched, so the capacity provider strategy remained simple.

Lab Steps

Step 1 – Verify the ECS Cluster

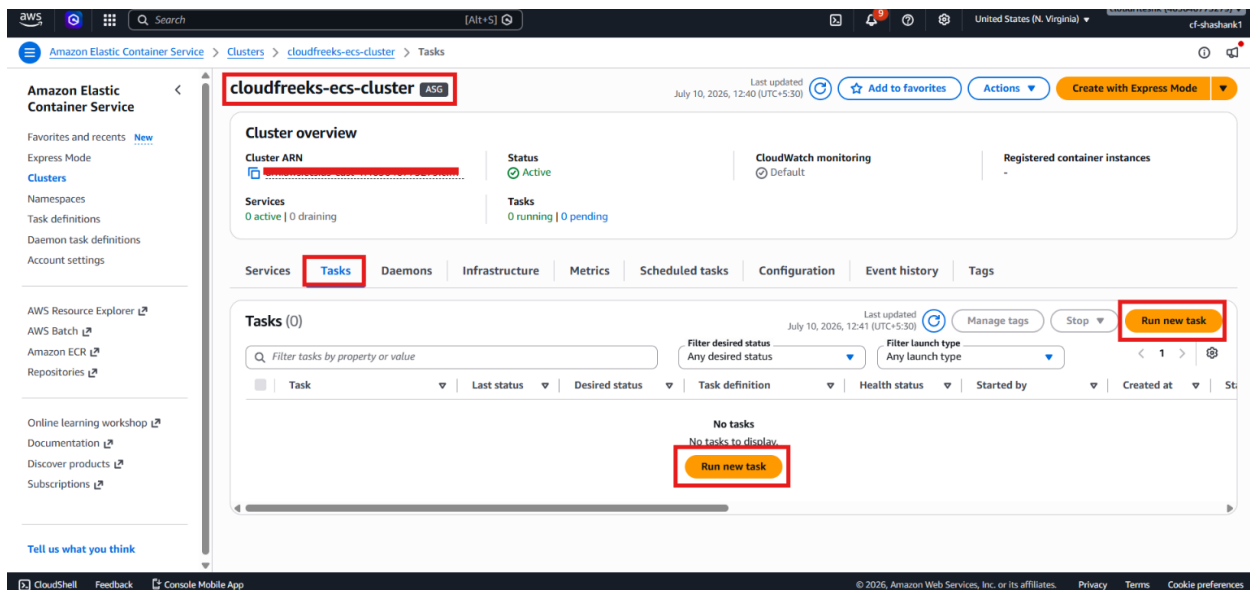
- Open the **Amazon ECS Console**.
- Confirm that the previously created ECS Cluster is available.



- If a cluster does not exist, create a new cluster with **AWS Fargate** enabled.

Step 2 – Run a New ECS Task

- Open the ECS Cluster.
- Select the **Tasks** tab.
- Click **Run new task**.



- Select the Task Definition Family created in the previous lab.

- Choose the latest Task Definition Revision.
- Set **Desired tasks** to 1.

Run task info

Task details

Task definition family
Select an existing task definition family. To create a new task definition, go to [Task definitions](#).

My-Web-App

Task definition revision latest
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

1

Desired tasks
Specify the number of tasks to launch.

1

Task group
All tasks with the same task group name are considered as a set when performing spread placement.

Environment Amazon EC2

Existing cluster
cloudfreeks-ecs-cluster

Compute configuration - advanced

Compute options Info

- Leave **Task Group** empty for the Fargate launch type.

Step 3 – Configure the Capacity Provider Strategy

- Choose **Custom Capacity Provider Strategy**.
- Select **FARGATE** as the capacity provider.
- Remove **FARGATE_SPOT** if only one Fargate task is required.
- Keep the default Base and Weight values.

Environment AWS Fargate

Existing cluster
cloudfreeks-ecs-cluster

Compute configuration - advanced

Compute options Info
To ensure task distribution across your compute types, use appropriate compute options.

☒ **Capacity provider strategy**
Capacity provider strategy to distribute your tasks across one or more capacity providers.

☐ **Launch type**
Launch tasks directly without the use of a capacity provider strategy.

Capacity provider strategy Info
Select either your cluster default capacity provider strategy or select the custom option to configure a different strategy.

☐ Use cluster default

☒ **Use custom (Advanced)**

Capacity provider
FARGATE

Base Info
0

Weight Info
1

Add capacity provider
You can add up to 2 more capacity provider strategy items.

Platform version Info
Specify the platform version on which to run your service.

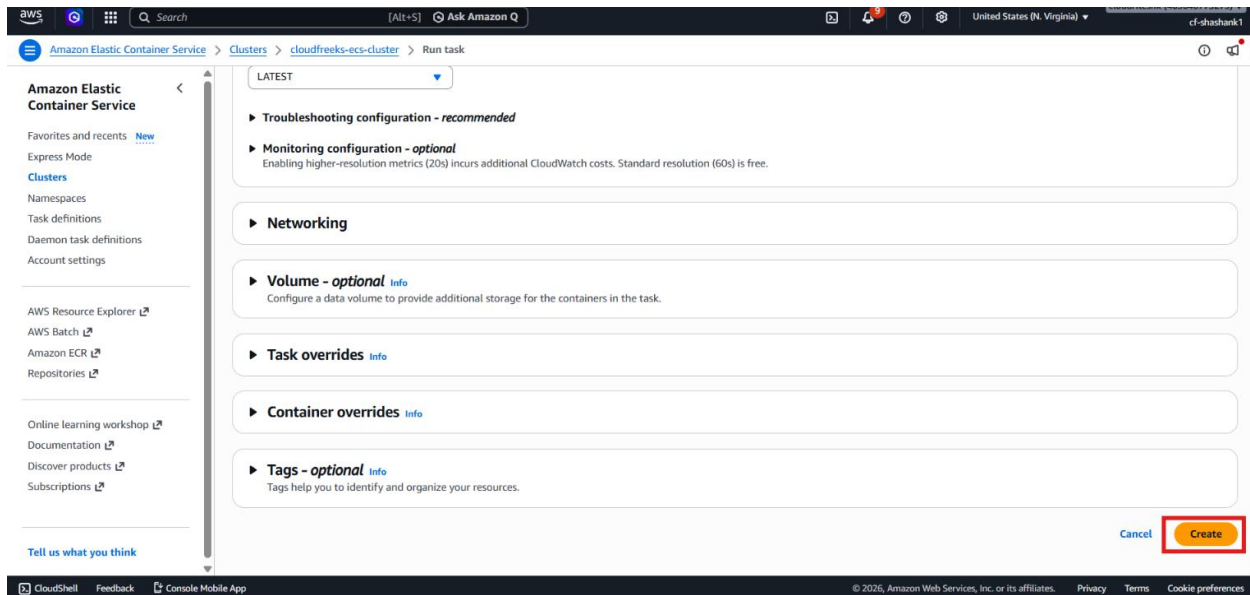
LATEST

Troubleshooting configuration - recommended

- Continue with the task creation.

Step 4 – Launch the Task

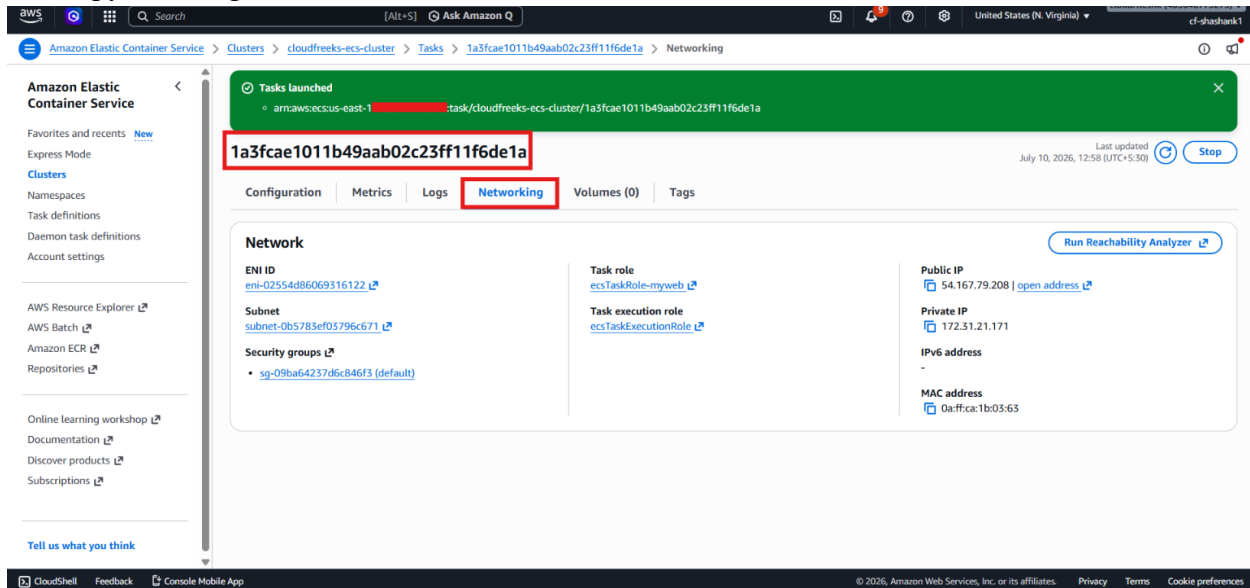
- Review all task settings.
- Click **Create**.



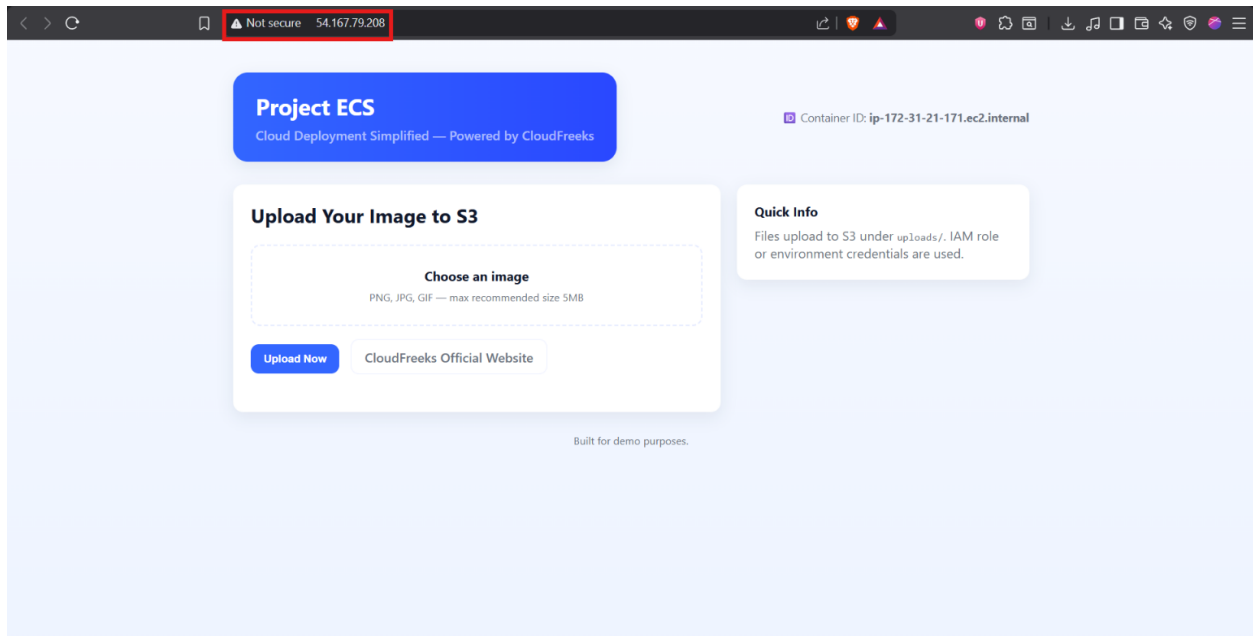
- Wait until the task status changes from **Provisioning** to **Running**.
- Open the task details after it starts successfully.

Step 5 – Test the Application

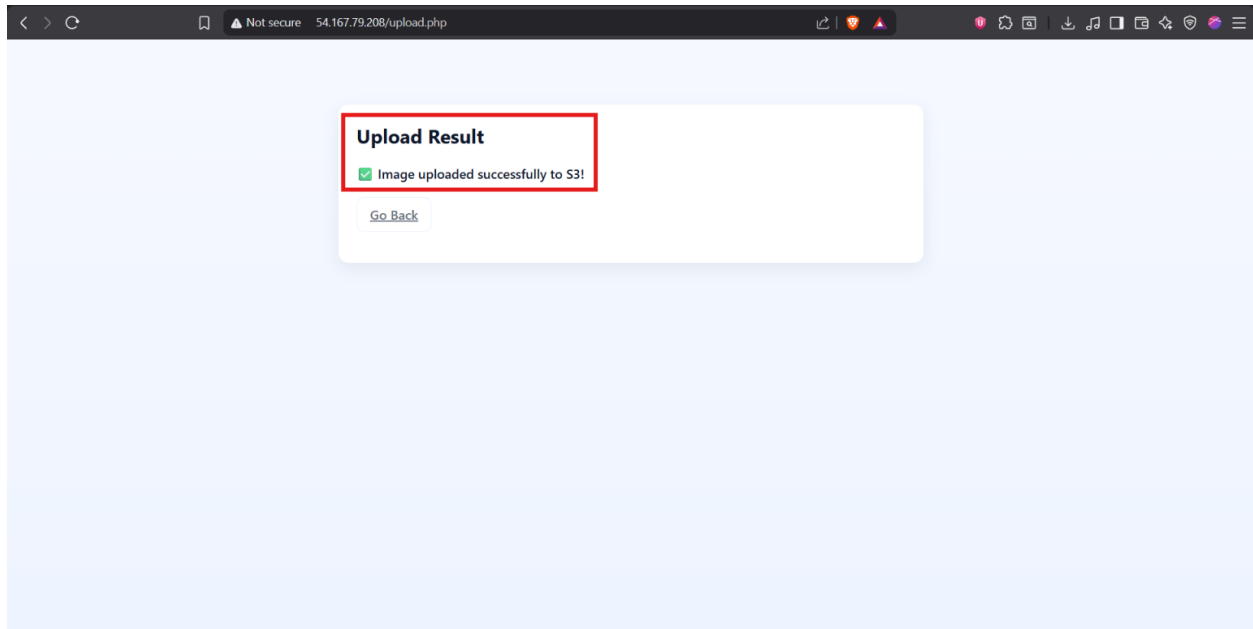
- Open the **Networking** section of the running task.
- Copy the assigned Public IP Address.



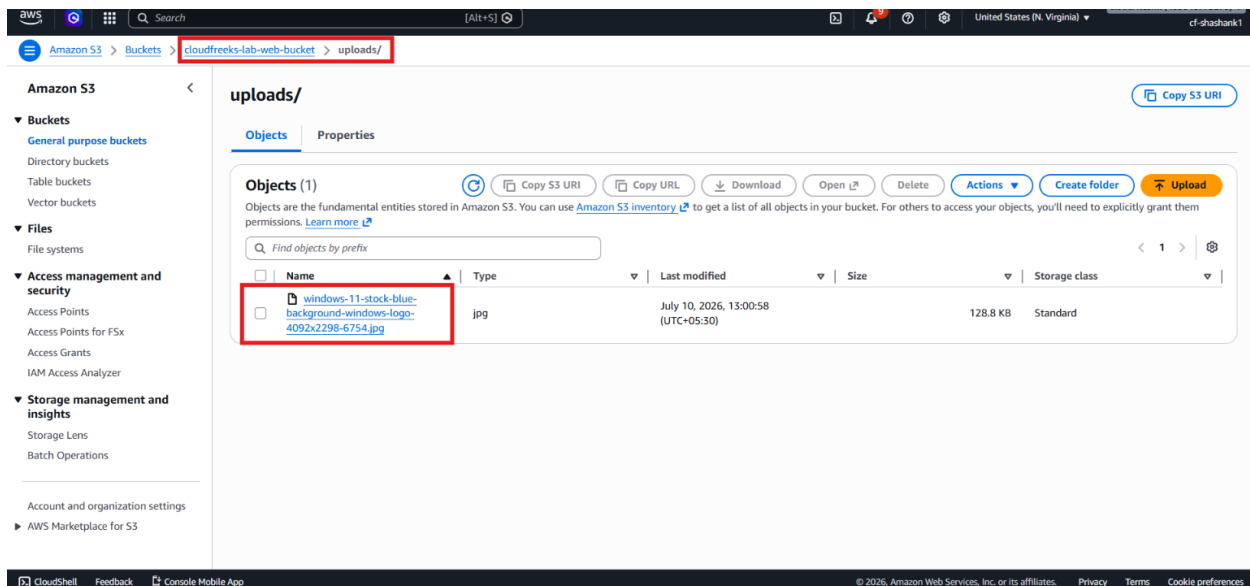
- Open the IP address in a browser using **HTTP**.
- Verify that the PHP application loads successfully.



- Upload a sample image.
- Confirm that the upload succeeds.

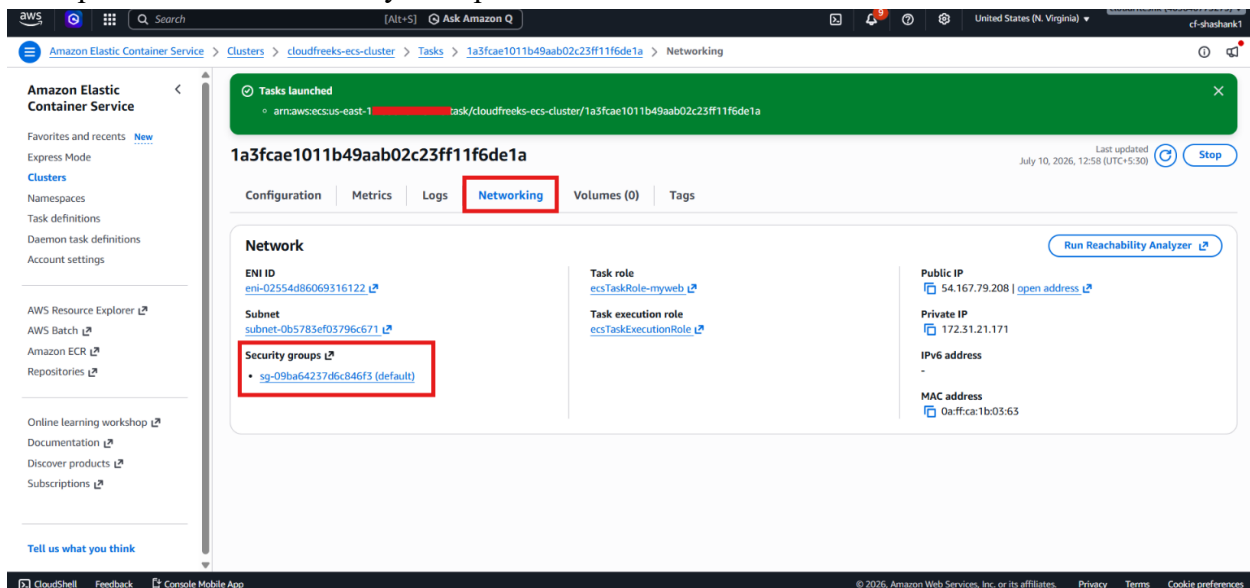


- Open the configured Amazon S3 bucket.
- Verify that the uploaded file is present inside the bucket.

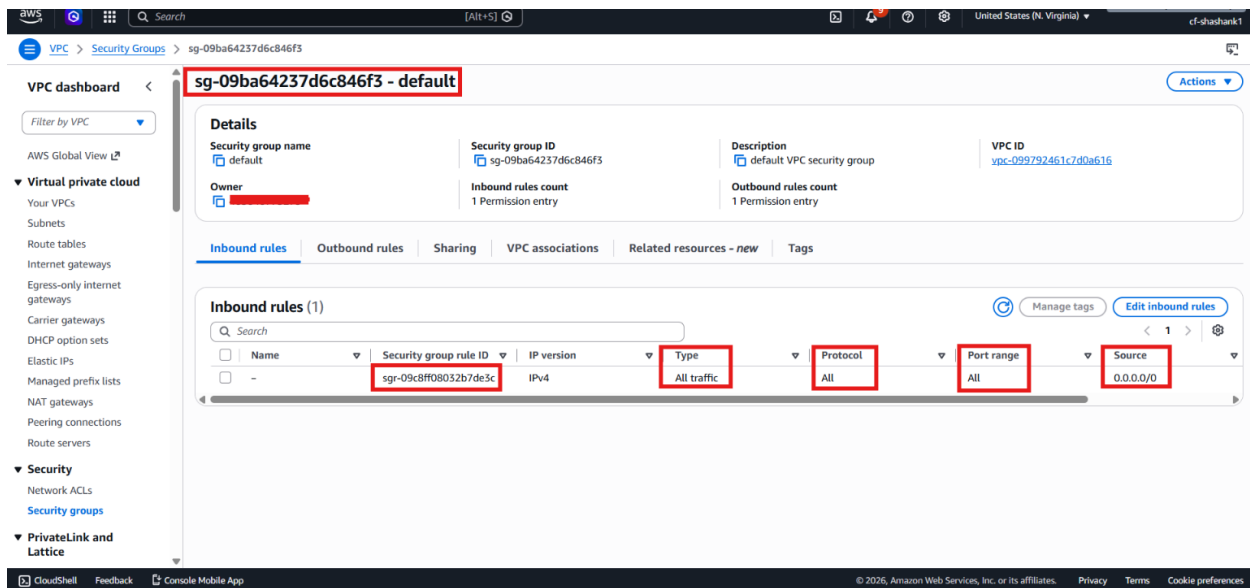


Step 6 – Verify Security Group (If Application Is Not Accessible)

- Open the running ECS Task.
- Navigate to **Networking**.
- Open the attached Security Group.



- Verify that the inbound rules allow **All Traffic** (or the required ports) from **0.0.0.0/0**.
- Save the updated security group if changes are required.



- Retry accessing the application.

Verification

- Verified that the ECS Cluster was available.
- Verified that the Task Definition Family and latest Revision were selected.
- Verified that the ECS Task entered the **Running** state.
- Verified that a Public IP Address was assigned to the task.
- Verified that the PHP application was accessible through the Public IP using **HTTP**.
- Verified that an image upload completed successfully.
- Verified that the uploaded image was stored successfully in the configured Amazon S3 bucket.
- Verified that the attached Security Group allowed the required inbound traffic for application access.